Week 06: Symbolic and Connectionist AI: Systematicity and Compositionality

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Two traditions have competed to explain intelligence: symbolic systems that manipulate structured representations and connectionist networks that learn distributed ones. This week examines their central dispute, whether networks can explain the systematicity of thought, and the empirical tests that now bear on it [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to contrast symbolic and connectionist architectures. They should be able to reconstruct the systematicity argument of Fodor and Pylyshyn and Smolensky's reply. They should also be able to explain why compositional generalisation tests such as SCAN bear on the argument. Finally, they should be able to evaluate recent evidence and neurosymbolic proposals.

Lecture

Two traditions

Symbolic AI treats intelligence as the manipulation of structured symbols by rules, in the tradition of the physical symbol system hypothesis from Week 3. Connectionism treats it as the result of learning in networks of simple units whose knowledge is stored in the strengths of their connections. The volumes on parallel distributed processing by Rumelhart, McClelland and colleagues made connectionism a serious rival in the 1980s by showing that networks could learn, generalise and degrade gracefully [4]. Present deep learning systems, including language models, descend from this tradition.

The systematicity argument

Fodor and Pylyshyn argued that human thought has two properties that a cognitive architecture must explain [1]. Thought is productive: People can think indefinitely many new thoughts.

Thought is systematic: Anyone who can think that John loves Mary can also think that Mary loves John. Classical architectures explain both properties, because their representations have a combinatorial syntax and semantics: The same parts are recombined by the same rules. Connectionist networks, they argued, face a dilemma. Either they implement a classical architecture, in which case they offer no new theory of cognition, or they do not, in which case they fail to explain systematicity.

Smolensky replied that connectionist models work at a subsymbolic level, below the level of concepts, and that structured representations can be encoded in distributed patterns of activity [2]. On his proposal, symbolic descriptions are approximations of the underlying dynamics rather than the mechanism itself. The dispute therefore concerns explanation as well as performance: whether systematicity is guaranteed by the architecture or merely possible after suitable training.

Empirical tests

Lake and Baroni turned the argument into an experiment [3]. Their SCAN benchmark maps commands such as jump twice and walk left to action sequences. Sequence-to-sequence networks generalised well when test commands were drawn at random. They failed when a primitive such as jump had been seen only in isolation during training and had to be combined with modifiers at test time. People, by contrast, generalise from a single example. Marcus used such results in a broader critique that listed the limitations of deep learning, including limited transfer and the absence of hierarchical structure [5].

Later work complicated the picture. Lake and Baroni showed that a standard neural network trained with meta-learning for compositionality, across many tasks that each required systematic generalisation, reached human-like performance on new instruction-learning tasks [6]. Systematicity, on this evidence, is achievable by networks but depends on how they are trained. Figure 6.1 reproduces the original contrast on a small command language.

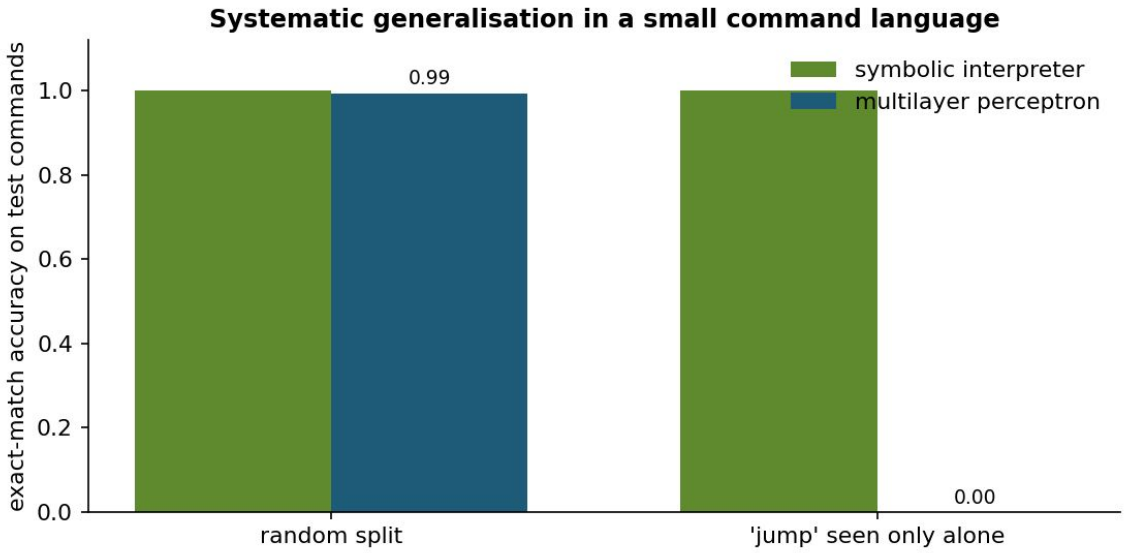


Figure 6.1. Exact-match accuracy of a symbolic interpreter and a multilayer perceptron on a random split and on a split in which the primitive jump appears only alone during training [3].

Hybrid prospects

d'Avila Garcez and Lamb described neurosymbolic AI as a third wave that integrates learning with reasoning over structured representations [7]. For philosophy of mind, the lesson of the debate is that systematicity has become an empirical and architecture-sensitive question. The interactive lab compares three systems on the same command language. They are a hand-written interpreter, a learner that memorises examples and a learner that acquires word meanings but composes them with fixed grammatical rules. The pattern of their successes and failures shows what compositional structure contributes.

> ***Pause and reflect.*** *If a network becomes systematic only after training on many systematic tasks, has it explained systematicity or inherited it from its training?*

Hands-on components

Interactive lab: A small command language maps commands such as jump twice and walk left to actions. Compare a memorising learner and a learner with compositional structure against the interpreter on different training splits, and type your own commands [1, 3]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-06/lab.html

Colab notebook: It generates all commands of the small language and trains a multilayer perceptron that predicts the action at every output position. It compares the exact-match accuracy of the network with a symbolic interpreter on a random split and on the held-out primitive split [1, 3]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-06/NB06_systematicity.ipynb

Reflection prompts

1. Which system in the interactive lab behaves most like a person learning a new verb? What does that suggest about the architecture of human cognition?
2. Does the success of meta-learning answer Fodor and Pylyshyn's dilemma, or does it choose its first horn?
3. Describe a compositional generalisation that matters in your field and how you would test a model for it.

Weekly task and submission

What to do. Evaluate the systematicity argument with evidence. First, state the dilemma of Fodor and Pylyshyn in two sentences. Second, compare the results of the Colab notebook with the SCAN results of Lake and Baroni. Use three splits: the random split, the split in which jump

is seen only alone and your own length split. Third, state which horn of the dilemma present neural networks fall under, if any.

Where to look. Lecture: sections "The systematicity argument" and "Empirical tests". Interactive lab: the comparison of the interpreter with two learners on different training splits (tab: Explore). Colab notebook: Section 3 with Exercise 1, and Exercise 2, where you design the length split.

Example. Figure 6.1 of the lecture shows the contrast that the Colab notebook reproduces. The network is almost perfect on the random split but fails on the commands that combine jump with a modifier. A student could argue that such a network fails to explain systematicity, and then ask whether the meta-learning results of Lake and Baroni change the verdict.

What to submit. About 500 words with a table of the accuracies on the three splits, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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[6] Lake, B. M., & Baroni, M. (2023). Human-like systematic generalization through a meta-learning neural network. Nature, 623(7985), 115-121. https://doi.org/10.1038/s41586-023-06668-3

[7] d'Avila Garcez, A., & Lamb, L. C. (2023). Neurosymbolic AI: The 3rd wave. Artificial Intelligence Review, 56(11), 12387-12406. https://doi.org/10.1007/s10462-023-10448-w